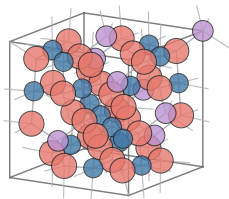


aexperimental MgAl_2O_4

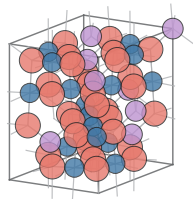
D1 uniaxial compression

D2 cation–cation swap

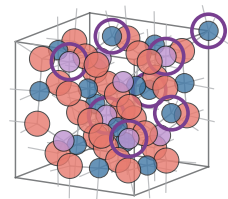
● Mg
● Al
● O



satisfies all laws
 $\rho = 0.99 \cdot \text{sites} = 0.21 \cdot \text{BV} = 0.04$



Law 1 $\rho = 0.76 \cdot \text{Law 8 BV} = 0.73$

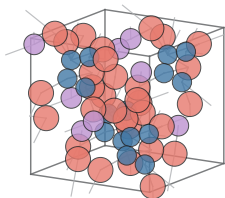


satisfies all laws
 $\rho = 0.92 \cdot \text{sites} = 0.57 \cdot \text{BV} = 0.10$

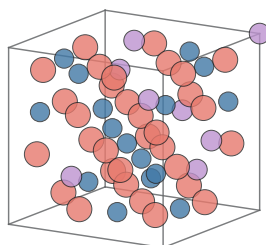
D3 random displacement

D4 isotropic expansion

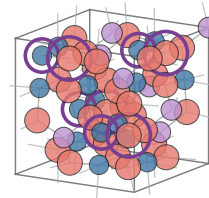
D5 cation–anion swap



Law 7 sites = 1.00 · Law 8 BV = 1.07
 Law 1 $\rho = 0.41 \cdot \text{Law 6 like bonds} = 0.47$



Law 2 $\rho = 1.28 \cdot \text{Law 8 BV} = 0.79$



Law 6 like bonds = 0.25 · Law 7 sites = 0.79

b

contact floor Law 1 alone (Set 1)	0.18	0.34	0.87	0.00	0.03
Law 2 ionic-contact law added to Set 1	0.00	0.00	0.00	0.48	0.01
Law 1 at 0.804 + Law 3–Law 5 added to Set 1'	0.38	0.20	0.07	0.26	0.22
like-charge ban Law 6 added to Set 2	0.06	0.01	0.01	0.01	0.31
distinct-site Law 7 added to Set 3	0.06	0.35	0.04	0.04	0.41
bond-valence Law 8 added to Set 3	0.07	0.04	0.00	0.17	0.03
Set 4 (seven laws)	0.73	0.91	1.00	0.93	0.99
	D1	D2	D3	D4	D5
	Damage type				

c